

How to Troubleshoot an AS2 Transmission Failure

A practical guide for diagnosing and resolving outbound AS2 transmission failures.

**Applies to**

AS2 / HTTP / HTTPS

**Transactions**

All X12 (e.g., 850)

**Task type**

Troubleshooting

**Estimated time**

15–30 minutes

Introduction

AS2 (Applicability Statement 2) is a secure method for exchanging EDI files over the internet using HTTP or HTTPS. During an AS2 transmission, your system connects to the trading partner's AS2 endpoint and sends the file. Depending on the trading partner agreement, the message is usually signed and encrypted, and the partner returns a Message Disposition Notification (MDN) that confirms the result.

This guide shows how to troubleshoot common outbound AS2 transmission failures and identify the root cause using logs, HTTP responses, and MDN details. The steps apply to any B2B integration platform that supports AS2.

Scope

**IN SCOPE**

- Outbound AS2 transmissions
- Connection, authentication, certificate, and MDN issues
- Any X12 transaction sent over AS2 (e.g., 850, 810, 856)

**NOT COVERED**

- AS1 or AS3
- FTP, SFTP, or other protocols
- EDI mapping or translation errors
- Trading partner onboarding procedures

Before You Start

Make sure you have the following:



The AS2 Message-ID, file name, and transmission timestamp of the failed message.



Access to your B2B integration platform's AS2 logs, transmission monitor, and AS2 configuration.



The trading partner's AS2 profile: endpoint URL, AS2 identifiers, certificates, and MDN settings.

Example Scenario

An outbound X12 850 to trading partner NORTHWINDSUP was generated and translated successfully. However, the AS2 transmission failed, and the monitoring system reported the following error:

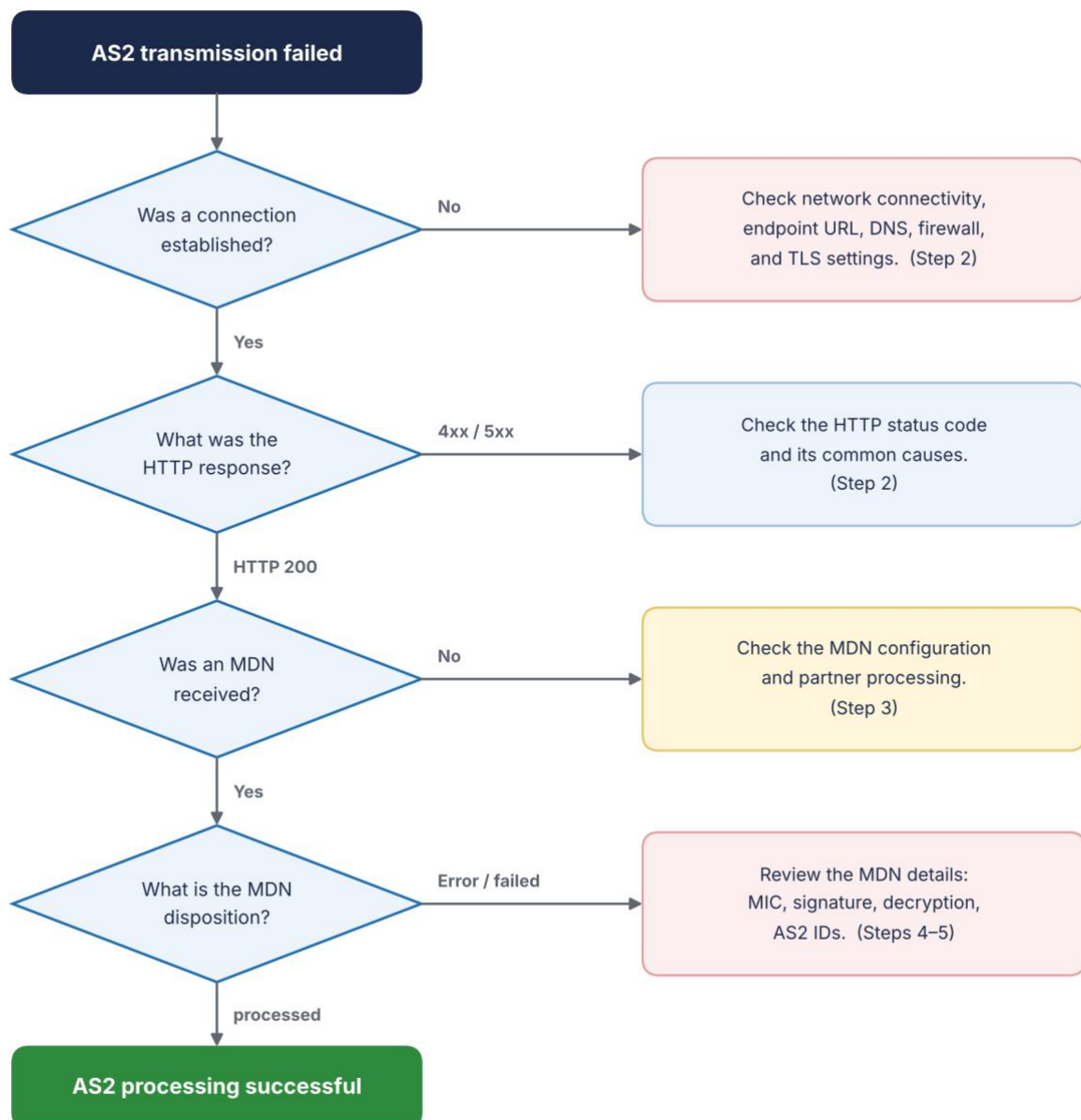
AS2 transmission failed: HTTP 403 Forbidden

The following message details are used in the main example. Steps 3 and 4 use separate examples for failures that occur after an HTTP 200 response.

Field	Value
AS2 Message-ID	<20261005150231.4521@acmeretail.example.com>
AS2-From	ACMERETAIL
AS2-To	NORTHWINDSUP
File name	850_ACMERETAIL_NORTHWINDSUP_4521.edi
Endpoint URL	https://as2.northwind.example.com:4080/as2/receive
Sent	2026-10-05 15:02:31

Troubleshooting Flow

Use this flowchart to decide where to start. Each outcome points to the step that covers it.



Troubleshooting Steps

Follow these steps to find the cause of the AS2 transmission failure.

1

Check the AS2 transmission status

Open your B2B integration platform's AS2 transmission monitoring interface and locate the failed message using its **AS2 Message-ID** or file name.

Item	What to check	Expected result
Message-ID	Unique AS2 Message-ID of the transmission	Matches the ID reported in the alert
File name	EDI file attached to the message	The correct 850 file was sent
AS2-From / AS2-To	Sender and receiver AS2 identifiers	Match the trading partner agreement
Timestamp	Date and time of the send attempt	Matches the expected send time
Status	Transmission status and error text	Shows the failure point and error message



Tip

If the platform retried the message, review the first failed attempt. Later retries can show a different error that hides the root cause.

2

Verify the connection and HTTP response

Open the transmission log for the message. Check whether a connection was made to the partner's AS2 endpoint. If it was, find the **HTTP status code** that the partner's server returned.

HTTP status	Meaning	Common causes
200	OK – request received. AS2 processing is not yet confirmed	Go to Step 3 and confirm that an MDN was received
401 / 403	Unauthorized / Forbidden	Credentials, IP allowlist, or partner access rules
404	Not Found	Incorrect endpoint URL or path
413	Payload Too Large	File exceeds the partner's size limit
500 / 503	Server error / Unavailable	Partner system issue or maintenance
No response	Connection failed	Network, DNS, firewall, or TLS issue

Example log excerpt (HTTP 403)

```
2026-10-05 15:02:31 INFO Connecting to https://as2.northwind.example.com:4080/as2/receive
2026-10-05 15:02:32 INFO TLS handshake completed
2026-10-05 15:02:32 INFO Sending message <20261005150231.4521@acmeretail.example.com>
2026-10-05 15:02:33 ERROR HTTP 403 Forbidden returned by partner endpoint
```

In the example scenario, the partner returned **HTTP 403 Forbidden**. The connection worked, but the partner's server refused the message.



Important

A 403 often means that the partner's firewall or IP allowlist does not include your outbound IP address. Confirm your current outbound IP with your network team before you contact the partner, especially after an infrastructure or cloud migration.

3

Confirm that an MDN was received

If the HTTP response is 200, check whether an MDN was received. The **MDN** is an acknowledgment returned by the trading partner that shows the AS2 processing result. Depending on the agreed configuration, the MDN may be signed or unsigned.

Item	What to check	Expected result
MDN mode	Synchronous or asynchronous, as agreed with the partner	Matches the partner's configuration
MDN received	Synchronous: in the HTTP response. Asynchronous: sent later to your MDN URL	MDN is linked to the original Message-ID
MDN URL	Asynchronous only: the return URL in the message headers	Reachable by the partner from the internet

MDN timing	Time between the send and the MDN receipt	Received within the agreed timeframe
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Additional Example A: HTTP 200, no MDN (separate transmission)

2026-10-06 09:15:44 INFO Sending message <20261006091544.4533@acmeretail.example.com>
 2026-10-06 09:15:46 INFO HTTP 200 OK (asynchronous MDN requested)
 2026-10-06 09:15:46 INFO Waiting for MDN
 2026-10-06 09:45:46 WARN No MDN received after 30 minutes

**Note**

With an asynchronous MDN, HTTP 200 only confirms that the partner's server accepted the request. It does not confirm that the partner processed the file.

4**Review the MDN disposition and MIC**

If an MDN was received, open it and review the **Disposition** field. If a **Received-content-MIC** is present, compare it with the MIC that your platform calculated when it sent the message. For signed MDNs, also verify the digital signature.

The disposition ends with a **disposition type**, an optional **modifier**, and a description. For example, in processed/error: decryption-failed, processed is the type, error is the modifier, and decryption-failed describes the error.

Type / modifier	Meaning	Action
processed	The partner received and processed the AS2 message	AS2 processing successful. Check the EDI functional acknowledgment (e.g., 997) separately, if required
processed/warning	Processed, but with a warning	Review the warning text and confirm with the partner
processed/error	The message was received, but processing failed (negative MDN)	Review the error description, then go to Step 5
failed/failure	The partner could not return the MDN as requested, for example because of an unsupported MIC or signature algorithm	Align the MDN signing and MIC algorithm settings with the partner

Additional Example B: Negative MDN (separate transmission)

Reporting-UA: NORTHWINDSUP AS2 Server
 Original-Recipient: rfc822; NORTHWINDSUP
 Final-Recipient: rfc822; NORTHWINDSUP
 Original-Message-ID: <20261007110207.4540@acmeretail.example.com>
 Disposition: automatic-action/MDN-sent-automatically; processed/error: decryption-failed

**Important**

A processed MDN with a MIC mismatch is not a clean success. The partner may have received an altered file. Treat it as a failure until you confirm the cause.

5**Investigate common AS2 issues**

Use the HTTP status, MDN disposition, and log details to find the cause. Check the following areas in your AS2 configuration and in the partner's AS2 profile. See **Appendix A** for a complete list of common errors.

Area	What to verify
Certificates	The partner's encryption certificate and your signing certificate are current, not expired, and match the certificates exchanged with the partner
Algorithms	Signing (e.g., SHA-256) and encryption (e.g., AES-256) algorithms match the partner agreement
AS2 identifiers	AS2-From and AS2-To values match exactly, including case and spaces
Endpoint	URL, port, and path match the partner's current AS2 profile
MDN settings	MDN mode, signed MDN option, and MIC algorithm match the partner agreement
Network	Firewall rules, IP allowlists, and TLS versions are accepted on both sides

i Tip

Certificate changes are the most common cause of sudden failures on a connection that worked before. Check whether either side recently renewed or replaced a certificate.

6**Determine next steps**

Based on your findings, take the appropriate action.

Scenario	Next step
Connection failure	Verify network, DNS, firewall, TLS, and the endpoint URL
HTTP 401 / 403	Check credentials, permissions, and the partner's IP allowlist
HTTP 404	Verify the AS2 endpoint URL and path with the partner
HTTP 500 / 503	Contact the trading partner and retry after they confirm recovery
No MDN received	Check the MDN configuration, or escalate to the partner
Negative MDN	Review the error details, correct the configuration, and resend

! Important

Before you resend, check the partner's processing status. If the partner already processed the file, do not resend. If the status is unknown, follow the duplicate-handling procedure agreed with the partner. A resend after a missing MDN can create a duplicate purchase order.

Result

You have identified the cause of the AS2 transmission failure and the recommended next steps.

Outcome	Example	Next action
Connection failed	Connection timeout	Verify network and endpoint settings
HTTP error	403 Forbidden	Check authentication and access rules
No MDN	HTTP 200, no MDN	Confirm the partner's MDN configuration
Negative MDN	processed/error: decryption-failed	Review certificates and correct the configuration
AS2 successful	processed	Check the EDI functional acknowledgment (e.g., 997), if required

i Tip

Record your findings, including the Message-ID, log excerpts, and MDN details. Partners resolve escalations faster when you include this evidence.

Appendix A: Common AS2 Errors

Use this table to match a symptom to its likely cause. Error text varies by platform, but the underlying causes are the same.

Symptom	Possible cause	Recommended action
Connection timeout	Network issue, or endpoint unavailable	Verify the URL, port, DNS, and firewall rules
TLS handshake failure	TLS version mismatch, or an untrusted or expired server certificate	Verify TLS settings and the partner's server certificate chain
HTTP 401 / 403	Authentication or access restriction	Check credentials, IP allowlist, and partner access rules
HTTP 404	Incorrect AS2 endpoint URL or path	Verify the partner URL
HTTP 413	File exceeds the partner's size limit	Enable compression, or agree on a size limit with the partner
HTTP 500 / 503	Partner system issue	Contact the trading partner
No MDN received	Partner processing delay, wrong MDN URL, or MDN delivery failure	Check the MDN configuration and partner logs
decryption-failed	Wrong or expired partner encryption certificate	Verify the encryption certificate and partner configuration
authentication-failed	Partner does not have your current signing certificate	Verify the signing certificate and send the current one to the partner
integrity-check-failed / MIC mismatch	Message altered in transit, or different MIC algorithms	Compare the returned MIC with the locally calculated MIC, and confirm the MIC algorithm
insufficient-message-security	Partner requires signing or encryption that was not applied	Enable the required signing or encryption options
unexpected-processing-error	Internal processing failure on the partner side	Contact the trading partner with the Message-ID
Invalid AS2 identifiers	Incorrect AS2-From or AS2-To values	Verify the AS2 identifiers in the message and partner configuration

MDN error descriptions such as *decryption-failed* and *authentication-failed* are defined in RFC 4130. Some platforms display them with different wording.



Need Help?

Found an issue or have a question?

Open an issue in the [edi-integration-playbook](#) repository on GitHub.

Document History

Version	Date	Author	Change
1.0	October 2026	alowinarp	Initial release